

BERGAMO ORIO AL SERIO, Italy

WMO: 160760

Lat: 45.669N

Lon: 9.700E

Elev: 778

StdP: 14.29

Time Zone: 1.00 (EUC)

Period: 1999-2023

Grade: A

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 25.2	(c) 27.8	(d) 9.4	(e) 9.1	(f) 35.9	(g) 13.9	(h) 11.5	(i) 35.4	(j) 16.8	(k) 51.2	(l) 13.7	(m) 49.3	(n) 3.7	(o) 10	(p) 0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.1	(c) 91.4	(d) 72.8	(e) 88.7	(f) 71.5	(g) 86.5	(h) 70.3	(i) 74.8	(j) 87.9	(k) 73.2	(l) 86.2	(m) 71.6	(n) 83.5	(o) 6.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.3	114.8	83.6	68.6	108.0	81.2	67.0	102.4	79.2	38.6	88.3	37.1	85.9	35.6	83.6	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.4	(b) 11.6	(c) 9.6	(e) 21.4	(f) 94.4	(g) 4.0	(h) 2.8	(i) 18.5	(j) 96.5	(k) 16.1	(l) 98.1	(m) 13.9	(n) 99.7	(o) 10.9	(p) 101.7		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			18.9	76.6	3.9	2.0	16.1	78.1	13.8	79.3	11.6	80.4	8.8	81.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures and Degree-Days	DBAvg	57.4	39.0	41.7	49.1	56.1	64.3	72.3	75.8	75.0	67.4	58.5	48.3	40.4
	DBStd	14.14	4.87	5.43	5.93	5.68	5.93	5.49	4.58	5.10	5.10	5.41	5.60	5.31
	HDD50	1075	343	234	86	13	0	0	0	0	4	95	300	
	HDD65	3826	807	653	494	271	87	7	0	1	33	209	501	763
	CDD50	3779	1	2	57	195	442	669	800	775	523	269	44	2
	CDD65	1054	0	0	0	3	64	226	335	311	106	9	0	0
(12) Wind	WSAvg	4.7	3.8	4.5	5.2	5.2	5.3	5.4	5.4	5.1	4.6	4.1	3.8	3.6
(13) Precipitation	PrecAvg	38.2	1.9	1.9	2.2	3.4	4.2	3.7	2.7	3.4	4.1	4.4	4.4	2.5
	PrecMax	67.4	9.8	7.3	7.3	7.3	8.7	11.2	7.8	7.8	14.5	10.3	13.7	7.6
	PrecMin	23.6	0.0	0.0	0.1	0.6	0.9	0.4	0.4	0.2	0.7	0.2	0.2	0.0
	PrecStd	10.1	2.0	1.9	1.7	1.6	2.1	2.5	1.9	2.0	3.1	2.5	3.5	1.9
(17) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	61.9	71.9	77.3	86.9	92.5	94.4	95.2	85.1	77.4	65.1	57.8
		MCWB	46.1	47.7	55.1	60.5	68.8	74.4	71.9	74.8	68.8	64.4	56.1	47.5
	2%	DB	52.4	56.8	66.6	73.4	82.4	89.4	91.1	91.5	82.6	73.2	61.3	53.5
		MCWB	44.4	46.6	52.0	58.1	66.1	71.5	72.5	72.8	67.2	61.9	54.4	46.5
	5%	DB	49.5	54.0	63.0	70.4	79.2	86.6	88.5	88.4	80.2	70.2	58.6	50.5
		MCWB	43.1	44.9	51.3	56.7	63.8	70.1	71.5	71.5	66.2	60.9	53.5	45.3
(25) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	46.7	51.0	59.7	67.4	75.9	83.9	86.3	85.7	77.5	67.3	56.2	48.1
		MCWB	42.0	43.5	49.6	55.2	62.2	68.3	70.3	70.3	64.4	59.4	52.2	44.4
	2%	WB	49.1	50.5	56.6	62.3	70.4	77.0	76.1	77.6	70.8	65.7	58.1	51.5
		MCDB	52.9	58.1	67.8	74.6	84.0	90.8	89.2	92.5	81.5	73.9	62.4	54.2
	5%	WB	47.0	48.4	54.0	59.7	67.3	73.6	74.3	75.0	69.2	64.0	56.2	48.6
		MCDB	49.8	54.0	63.3	70.5	79.9	86.7	87.4	88.1	79.5	70.8	59.4	50.8
(33) Mean Daily Temperature Range	5% DB	WB	45.0	46.8	52.4	58.0	65.2	71.3	73.0	73.1	67.8	62.3	54.7	46.8
		MCDB	47.3	51.4	60.5	68.0	76.4	83.7	85.9	85.9	77.2	68.3	57.5	49.0
	5% WB	WB	42.9	45.1	50.8	56.4	63.4	69.6	71.6	71.3	66.2	60.8	53.0	45.0
		MCDB	45.8	49.1	58.1	65.1	74.0	82.2	83.5	82.8	75.0	66.1	55.7	47.2
(38) Clear-Sky Solar Irradiance	taub	taub	0.315	0.349	0.395	0.455	0.463	0.488	0.469	0.459	0.446	0.416	0.361	0.315
		taud	2.494	2.374	2.230	2.066	2.096	2.075	2.153	2.176	2.182	2.231	2.399	2.509
	Ebn at Noon	Ebn at Noon	251	260	261	254	256	250	253	250	241	229	227	238
		Edh at Noon	21	29	38	49	49	50	46	43	40	33	23	19
(42) All-Sky Solar Radiation	RadAvg	RadAvg	477	746	1177	1479	1728	1984	2053	1732	1305	805	475	381
	RadStd	RadStd	62	102	132	168	144	124	139	110	89	105	75	54

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(44) Station	Trend	(e) N/S	(f) +1.71	(g) N/S	(h) N/S	(i) N/S	(j) N/S	(k) N/S	(l) N/S	(m) N/S	(n) N/S
(45) Station	Variability	1.3	2.5	3.2	2.2	1.8	2.2	229	315	317	218
(46) Regional Trend (22 neighbors)		+1.04	+1.93	N/S	+1.58	N/S	N/S	-194	-261	+185	+146

Nomenclature: See separate page